

INTERSECONDARY SCHOOLS EXAMINATION SERIES ISESE



FORM THREE SERIES No.01 PHYSICS 1

TIME 3 HRS

Year 2026

Instructions

1. This paper consists of sections A, B and C with a total of eleven (11) questions.
2. Answer all questions in section A and B and only two (2) questions from section C.
3. Non-programmable calculators may be used.
4. Where necessary the following constants may be used;
Acceleration due to gravity, $g = 10\text{m/s}^2$ or 10N/kg
Density of water = 1000kg/m^3
1 litre = $1\text{dm}^3 = 1000\text{cm}^3$
Specific heat capacity of water, $c = 4200\text{ J/kgK}$
Pie, $\pi = 3.14$

GEPAM science hub

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FORM ONE AND FORM THREE SERIES EXAMINATION ISESE TIME TABLE FOR REGISTRATION CALL US 0624 254 757

ALL SUBJECTS	DATE
SERIES - 01	03 rd August 2026
SERIES - 02	10 th August 2026
SERIES - 03	17 th August 2026
SERIES - 04	24 th August 2026
SERIES - 05	31 st August 2026

NB# SERIES No.06 – No.10 (SEPTEMBER – OCTOBER)

WE ALSO HAVE EXAMINATION SERIES FOR FORM TWO, FORM FOUR , FORM FIVE AND FORM SIX

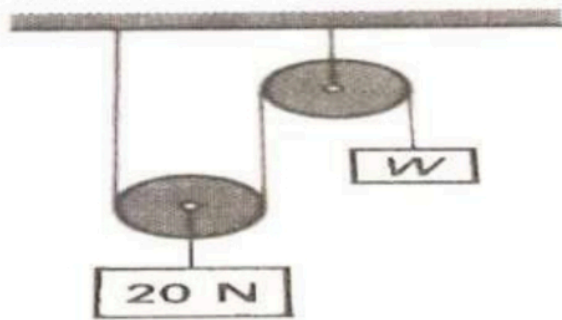
SECTION A (16 marks)

Answer ALL questions

1. For each of the items (i) – (x), choose the correct answer among the given alternatives and write its letter
 - i) The meniscus of mercury in glass vessels curves upward because
 - A. The cohesion of molecules of mercury is stronger than adhesion of molecules of a glass and mercury
 - B. The adhesion of molecule of molecules of mercury is stronger than the cohesion of molecules of mercury
 - C. Mercury experience downward force equal to its weights
 - D. Mercury is denser than glass
 - E. Glass is denser than mercury
 - ii) Steam at 100°C has more burning effect than water at 100°C because:
 - A. The latent heat of vaporization is higher than the specific heat capacity of water at 100°C
 - B. Steam is less dense than water
 - C. The molecules of water in steam have more kinetic energy than those of water at 100°C
 - D. Steam is at higher temperature than water at 100°C
 - E. None of them above
 - iii) If one cell in a three cell torch is placed in the opposite direction:
 - A. The torch will not give light
 - B. The torch will give very bright light.
 - C. The torch will give very dim light
 - D. The torch will give normal light
 - E. The torch will give dark light
 - iv) If Elisha Kipanga wants to establish a temperature scale then he should know the following:
 - A. Fixed points fundamental interval, an equation relating physical quantity and temperature change
 - B. Thermometer, ice point and steam point
 - C. Water, ice point mercury
 - D. Fundamental interval and thermometer
 - E. Water, mercury and thermometer
 - v) A metal rod has a length of 40cm on a day when the temperature of the room is 22.3°C. What will be its length on a day when the temperature of the room is 30°C and the linear expansivity of the metal is $0.0000017^{\circ}\text{C}^{-1}$?
A. 40.021cm B. 40.0052 cm C. 0.0052 cm D. 38.002 cm E. 39.994 cm
 - vi) A substance of mass 2kg is supplied with heat of 4800J, and causes its temperature to rise by 4K. The specific heat capacity of the substance is:
A. $0.6\text{JKg}^{-1}\text{K}^{-1}$ B. $600\text{JKg}^{-1}\text{K}^{-1}$ C. $38.4\text{JKg}^{-1}\text{K}^{-1}$
D. $2400\text{JKg}^{-1}\text{K}^{-1}$ E. $100\text{JKg}^{-1}\text{K}^{-1}$

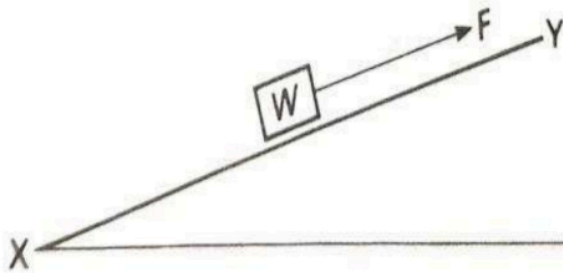
- vii) Increase in pressure raises the boiling point of a liquid. This principle is used in
- | | |
|-------------------------------------|-------------------------------------|
| A. Refrigerator and vacuum pump | B. Pressure cooker and oven |
| C. Pressure cooker and steam engine | D. Refrigerator and pressure cooker |
| E. Oven and iron | |

- viii) In figure below a weight W just supports the 20N weight. The value of W is

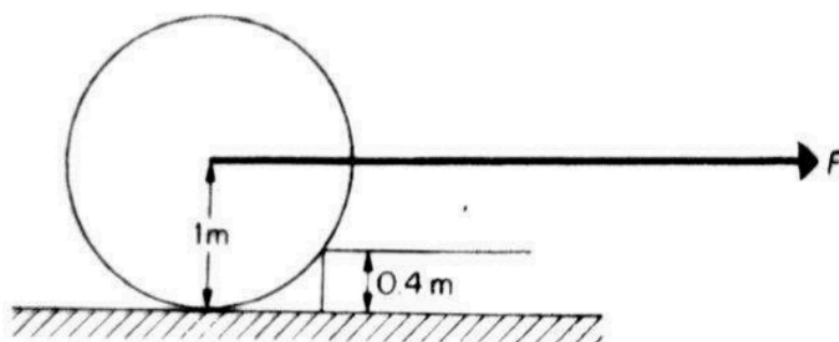


- | | | | | |
|-------|---------|--------|--------|-------|
| A. 2N | B. 6.7N | C. 10N | D. 20N | E. 5N |
|-------|---------|--------|--------|-------|

- ix) In the figure below a weight w pulled steadily a smooth inclined plane XY by a force. The mechanical advantage depends on the magnitude of:



- | | |
|----------------|---|
| A. W | B. F |
| C. Length XY | D. The angle XY makes with the horizontal |
| E. The height | |
- x) The diagram below show a wheel mass 15 kg and radius 1m being pulled by a horizontal force F against a step 0.4 m high . What initial force is just sufficient to turn the wheel so that it can raise over the step?



- A. 600N B. 60N C. 20N D. 200N E. 30N

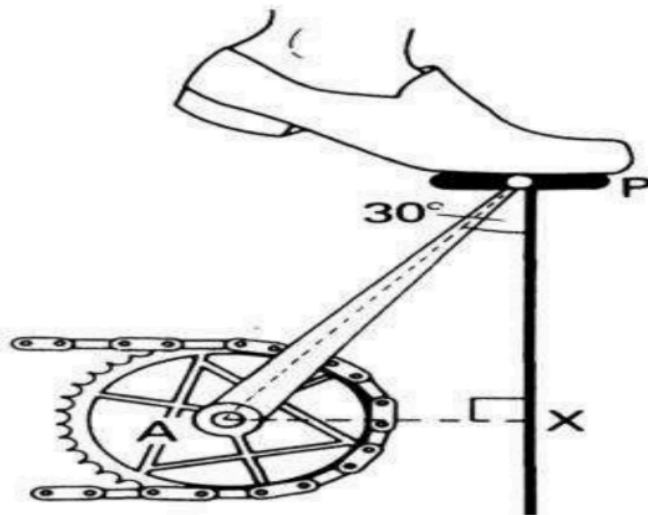
2. Match the parts of the diagram in **List A** with a correct response in **List B** by writing a letter response below the number of the corresponding item in **List A** in the table provided.

LIST A	LIST B
	A. Mixture of carbon and manganese (IV)oxide B. Carbon rod (+) C. Ammonium chloride solution D. Porous pot E. Zinc rod (-) F. Glass jar G. Mixture of powdered carbon and manganese (IV) oxide H. Ammonium chloride jelly

SECTION B (54 marks)

Answer **ALL** questions

3. (a) With your understand give the meaning of the term prism(2 marks)
(b) Explain two types of prisms(4 marks)
(c) Draw the graph which shows the variation of the angle of deviation with angle incidence.(3 marks)
4. (a) Explain factors affecting moment of force(3 marks)
(b) Suppose the pedal P of a bicycle slips on its bearing and becomes horizontally to the ground. Find the moment about the hinge at A
(i) If the downward vertical force on the pedal is 20N(3 marks)
(ii) If the crank AP is 15cm long and inclined at 30°Cto the vertical(3 marks)

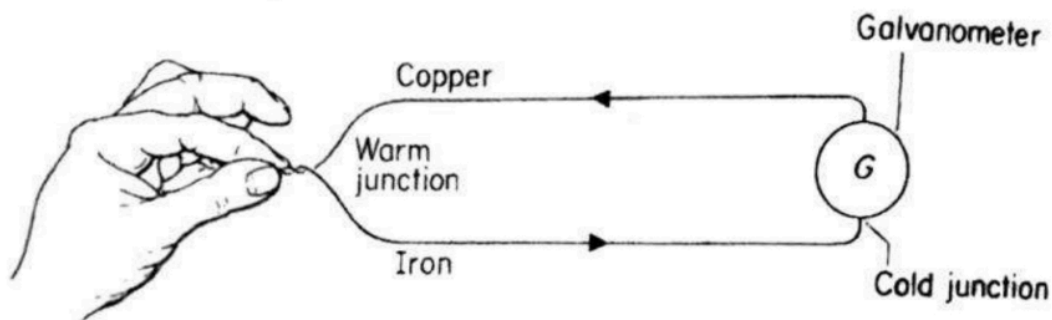


5. (a) State five laws of friction(**5 marks**)

(b) A weight of 20N is on an inclined plane at 40° to the horizontal. What are the components of the weight parallel and perpendicular to the plane?(**4 marks**)

6. (a) How can we detect radiant energy from hot body(**3marks**)

(b) A simple experiment below serves to show how this is done. A copper and an iron wire are twisted together to form a junction, while the free ends of the wires are connected to the terminals of a sensitive galvanometer.



On warming the junction an (i) _____ is produced in the circuit and the (ii) _____ gives a deflection. This is called (iii) _____.(**6 marks**)

7. (a) Why is water used as a coolant in car engines? (**4 marks**)

(b) Dry steam is passed into a well-lagged copper can of a mass 250g containing 400g of water and 50g of ice at 0°C . The mixture is well stirred and steam supply cut off when the temperature of the can and its contents reaches 20°C . Neglecting heat losses, find the mass of steam condensed.

(Specific heat capacities: water, 4.2J/gK ; Copper, 0.4J/gK .

Specific latent heats: steam, 2260J/g ; ice, 336J/g) (**5 marks**)

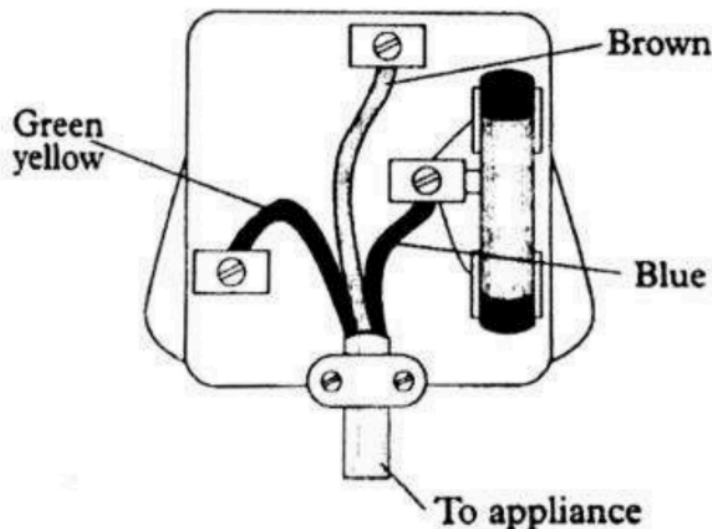
8. (a) Madam Angela Is an IT at U.B.N Company, she was trouble shouting a laptop mother board by using a multimeter and unfortunately, she discovered that an electronic device labelled 20F has some energy ;draw its electric symbol of this device and give its uses.(**4 marks**)

(b) Assume that your father does not know the mode of action of lighting conductor. With clear labelled diagram explain the mode of action of lighting conductor to him. **(5 marks)**

SECTION C (30Marks)

Answer **ONLY TWO** questions from this section

9. (a) The figure below shows a flex to the 13A – 3 pin plug which has been incorrectly fitted.



List the mistakes and suggest the corresponding remedies **(6 marks)**

(b) What would happen if the plug was connected to the mains sockets? **(4 marks)**

(c) A cell supplies a current of 0.6A through a 2 Ohms resistor and current of 0.2A through a 7 Ohms resistor. Calculate the e.m.f and the internal resistance of the cell. **(5 marks)**

10. (a) The velocity of a car A relative to a car B is 5m/s when the two cars are moving in the same direction and 15m/s when the two cars are moving in the opposite directions. Determine the velocity of each car **(10 marks)**

(b) A projection lantern is used to project a slide measuring 3cm × 3cm onto a screen 12cm from the projection lens. If the size of the screen is 1.5m × 1.5 m, how far from the lens must the slide be for the image to fill the entire screen? **(5 marks)**

11. (a) Why electrical cables are left sagging during installation on the hot day and taut during cold? **(4 marks)**

(b) A brass plate has a hole whose radius is too small for an iron rivet can fit in. Explain way that the rivet can be made to fit in the hole. **(5 marks)**

(c) A fixed mass of gas occupies a volume of $1.0 \times 10^{-3} \text{ m}^3$ at a pressure of 76cmHg. What's volume does the occupy at 17°C if it's pressure is 72cmHg? **(6 marks)**